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# Electromyographic activity of masticatory, neck and trunk muscles of subjects with different skeletal facial morphology – a cross-sectional evaluation

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SUMMARY **The electromyographic pattern activity of masticatory, neck and trunk muscles was assessed using surface electromyography (sEMG) in 60 Caucasian adult females (20 subjects in skeletal class I, 20 subjects in skeletal class II and 20 subjects in skeletal class III), classified on the base of their skeletal class (ANB angle), corrected on the base of maxillary and mandibular rotations. The sEMG activity was recorded at mandibular rest position and during maximal voluntary clenching. At mandibular rest position, the sEMG activities of masseter and anterior temporal muscles were significantly higher in class III subjects than in class I and class II subjects, that showed no significant difference between them. Then, the sEMG activities of posterior cervicals and upper trapezius were significantly higher in skeletal class III subjects than in the other two groups. During maximal voluntary clenching, no significant difference was observed in the sEMG activity of masticatory muscles among the three considered groups. However, the sEMG activities of posterior cervicals and upper trapezius were significantly higher in skeletal class III subjects than in the other two groups, which showed no significant difference between them. In conclusion, the skeletal class seems to affect the sEMG pattern activity of masticatory, neck and trunk muscles.**

KEYWORDS: **skeletal facial type, cephalometry, electromyography, masticatory muscles, neck and trunk muscles**

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## Introduction

In 1991, Miralles *et al.* (1) investigated the surface electromyographic (sEMG) activity of masticatory muscles in skeletal classes I, II and III subjects, classified on the base of ANB angle, corrected for maxillary and mandibular rotations. They found that skeletal class III subjects showed a significantly higher sEMG activity of anterior temporal and masseter than skeletal classes I and II subjects that showed no significant difference between them; on the contrary, during maximal voluntary clenching (MVC), no significant difference was observed in the sEMG activity of the two muscles among the three groups.

In this study, the authors investigated the sEMG activity of neck and trunk muscles in subjects classified according to Miralles *et al.* (1). The importance of this study concerns different aspects.

From a functional point of view, the sEMG activities of masticatory, neck and trunk muscles are strongly linked, due to a reciprocal innervation between the trigeminal and the cervical systems that produces mutual inhibition and activation; it seems to be present a dynamic relationship among dental occlusion, 'space condition' and head posture (2, 3).

The anatomical bases of these correlations were studied in some mammals, which showed the presence of neuronal connections between trigeminal afferents and cervical spinal cord (4): trigeminal sensory afferents were found to project in several 'non-trigeminal' areas of central nervous system, even in the lower cervical neuromers (5). Thus, trigeminal inputs from periodontal area, temporomandibular joint and muscular receptors could play some role in the modulation of cervical muscles.

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In dental literature, according to these concepts, several investigations were performed; for example, it was showed that voluntary teeth clenching normally provokes a co-activation of the sternocleidomastoid (SCM) muscle (6, 7).

In 1995, Zuniga *et al.* (4) found that the variation in occlusal contact influenced the sEMG activity of cervical muscles in 20 patients with myogenic cranio-cervical-mandibular dysfunction. They found that the higher sEMG activity of cervical muscles was associated to the MVC in the retrusive position.

As seen, the relationship between stomatognathic system and neck area seems to be well accepted in literature; for this, our hypothesis is that facial morphology, *id est* skeletal class, may be correlated to different patterns of sEMG activity of masticatory muscles, and also the neck and trunk muscles.

The aim of this investigation was to record and compare the sEMG activity of masticatory, neck and trunk muscles during different functional activities of stomatognathic system, in an adult sample, classified according to the corrected ANB angle.

## Subjects and methods

### *The sample*

The subjects included in the present investigation were selected from patients of the Department of Oral Science, Orthodontic Unit, University G.d'Annunzio, Chieti/Pescara, Italy.

The sample included 60 Caucasian adult females (mean age: 31·6 ± 4·8 years): 20 skeletal class I subjects (27·4 ± 3·2 years), 20 skeletal class II subjects (31·6 ± 5·5 years) and 20 skeletal class III subjects (32·2 ± 5·2 years). An inclusion criteria for the sample was the presence of all posterior teeth, until the second molars, without any dyrect and/or prosthetic restauration. All the patients had no complaints of temporo-mandibular joint disorders or muscular disorders and no history of bruxism. All the teeth were in occlusion; all the subjects showed the same dental occlusion of the skeletal class: skeletal class I subjects showed molar Angle's class I; skeletal class II subjects showed molar Angle's class II and skeletal class III subjects showed molar Angle's class III. Informed consents were obtained by all the subjects.

### *Cephalometric tracings*

All the cephalometric tracings were done by the same author (S.T.). Angular measurements used to classify the subjects were carried out on lateral skull radiographs (Fig. 1) by the same author (S.T.). The procedure was the same used by Miralles *et al.* (1): ANB angle was measured according to Steiner (8) and the values were adjusted according to Freeman (9), subtracting or adding 1° to the ANB value for every 2° that the SNA angle exceeded or was under 81·5°, respectively. Then, the values adjusted according to Freeman, were modified according to SN-GoGn angle (10). This procedure consisted of subtracting or adding 25% of the difference by which the SN-GoGn angle exceeded or was under 32°, respectively. Thus, original ANB angle, adjusted according to maxillary position and inclination of the jaws, was used to classify the subjects into three groups (Tables 1–3). When the value varied from 0° to 4°, subjects were considered in skeletal class I (Table 1). Subjects with values higher than 4° were classified in skeletal class II (Table 2), and subjects with negative values were considered in skeletal class III (Table 3).

On 10 randomly selected radiographs, the measurements were repeated after 1 week by the same author

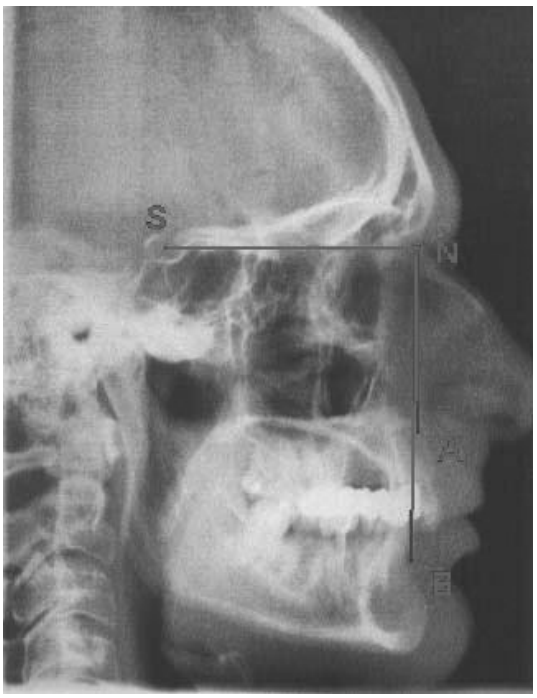


**Fig. 1.** Lateral skull radiographs.

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**Table 1.** Cephalometric variables on lateral skull radiographs of subjects in skeletal class I. See text for details

| Patient number | SNA (°) | SNB (°) | ANB (°) (Steiner) | ANB (°) (Freeman) | SN-GoGn (°) | Corrected ANB (°) | Skeletal class |
|---|---|---|---|---|---|---|---|
| 1 | 78·00 | 78·00 | 0·00 | 1·75 | 31·00 | 2·00 | I |
| 2 | 78·50 | 74·50 | 4·00 | 5·50 | 42·00 | 3·00 | I |
| 3 | 82·00 | 79·00 | 3·00 | 2·75 | 31·00 | 3·00 | I |
| 4 | 74·00 | 73·00 | 1·00 | 4·75 | 35·00 | 4·00 | I |
| 5 | 79·00 | 81·50 | −2·50 | −1·25 | 22·00 | 1·25 | I |
| 6 | 80·00 | 80·00 | 0·00 | 0·75 | 26·00 | 2·25 | I |
| 7 | 80·00 | 75·00 | 5·00 | 5·75 | 41·00 | 3·50 | I |
| 8 | 72·50 | 75·00 | −2·50 | 2·00 | 30·00 | 2·50 | I |
| 9 | 80·00 | 77·00 | 3·00 | 3·75 | 36·00 | 2·75 | I |
| 10 | 80·50 | 77·00 | 3·50 | 4·00 | 41·00 | 1·75 | I |
| 11 | 81·00 | 78·50 | 2·50 | 2·75 | 39·00 | 1·00 | I |
| 12 | 72·00 | 73·00 | −1·00 | 3·75 | 39·00 | 2·00 | I |
| 13 | 78·00 | 75·50 | 2·50 | 4·25 | 34·00 | 3·75 | I |
| 14 | 81·50 | 78·00 | 3·50 | 3·50 | 30·00 | 4·00 | I |
| 15 | 84·00 | 82·00 | 2·00 | 0·75 | 33·00 | 0·50 | I |
| 16 | 81·00 | 80·00 | 1·00 | 1·25 | 32·00 | 1·25 | I |
| 17 | 80·00 | 78·00 | 2·00 | 2·75 | 33·00 | 2·50 | I |
| 18 | 82·00 | 80·00 | 2·00 | 1·75 | 33·00 | 1·50 | I |
| 19 | 84·00 | 80·00 | 4·00 | 2·75 | 34·00 | 2·25 | I |
| 20 | 84·00 | 82·00 | 2·00 | 0·75 | 32·00 | 0·75 | I |

**Table 2.** Cephalometric variables on lateral skull radiographs of subjects in skeletal class II. See text for details

| Patient number | SNA (°) | SNB (°) | ANB (°) (Steiner) | ANB (°) (Freeman) | SN-GoGn (°) | Corrected ANB (°) | Skeletal class |
|---|---|---|---|---|---|---|---|
| 1 | 79·50 | 75·00 | 4·50 | 5·50 | 33·00 | 5·25 | II |
| 2 | 76·50 | 73·00 | 3·50 | 6·00 | 36·00 | 5·00 | II |
| 3 | 78·50 | 71·00 | 7·50 | 9·00 | 47·00 | 5·25 | II |
| 3 | 76·50 | 71·50 | 5·00 | 7·50 | 30·00 | 8·00 | II |
| 4 | 75·00 | 71·00 | 4·00 | 7·25 | 42·00 | 4·75 | II |
| 5 | 79·00 | 74·00 | 5·00 | 6·25 | 34·00 | 5·75 | II |
| 6 | 80·00 | 74·00 | 6·00 | 6·75 | 37·00 | 5·50 | II |
| 7 | 83·00 | 79·00 | 4·00 | 3·25 | 23·00 | 5·50 | II |
| 8 | 83·00 | 78·00 | 5·00 | 4·25 | 29·00 | 5·00 | II |
| 9 | 85·00 | 78·50 | 7·50 | 5·75 | 24·00 | 7·75 | II |
| 10 | 72·00 | 63·50 | 8·50 | 13·25 | 40·00 | 11·25 | II |
| 11 | 81·00 | 76·00 | 5·00 | 5·25 | 36·00 | 4·25 | II |
| 12 | 79·50 | 77·50 | 2·00 | 3·00 | 27·00 | 4·25 | II |
| 13 | 82·20 | 77·90 | 4·30 | 3·95 | 24·00 | 5·95 | II |
| 14 | 87·00 | 82·00 | 5·00 | 2·25 | 24·00 | 4·25 | II |
| 15 | 84·00 | 78·00 | 6·00 | 4·75 | 33·00 | 4·50 | II |
| 16 | 84·00 | 77·00 | 7·00 | 5·75 | 34·00 | 5·25 | II |
| 17 | 85·00 | 80·00 | 5·00 | 3·25 | 28·00 | 4·25 | II |
| 18 | 85·50 | 80·50 | 5·00 | 3·00 | 26·00 | 4·50 | II |
| 19 | 86·00 | 81·00 | 5·00 | 2·75 | 26·00 | 4·25 | II |
| 20 | 85·00 | 80·00 | 5·00 | 3·25 | 28·00 | 4·25 | II |

(S.T.) and compared with the first ones, using the Dahlberg's formula (11), to evaluate the method error.

$$\delta = \sqrt{(\sum d^2/2N)},$$

where $d$ is the difference between the first and the second measurements and $N$ is the number of double registrations.

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**Table 3.** Cephalometric variables on lateral skull radiographs of subjects in skeletal class III. See text for details

| Patient number | SNA (°) | SNB (°) | ANB (°) (Steiner) | ANB (°) (Freeman) | SN-GoGn (°) | Corrected ANB (°) | Skeletal class |
|---|---|---|---|---|---|---|---|
| 1 | 81·00 | 83·00 | −2·00 | −1·75 | 26·00 | −0·25 | III |
| 2 | 77·00 | 78·00 | −1·00 | 1·25 | 38·00 | −0·25 | III |
| 3 | 75·00 | 80·00 | −5·00 | −1·75 | 26·50 | −0·37 | III |
| 3 | 72·00 | 74·00 | −2·00 | 2·75 | 44·00 | −0·25 | III |
| 4 | 82·00 | 82·00 | 0·00 | −0·25 | 33·00 | −0·50 | III |
| 5 | 89·00 | 90·00 | −1·00 | −4·75 | 25·00 | −3·00 | III |
| 6 | 84·00 | 86·00 | −2·00 | −3·25 | 26·00 | −1·75 | III |
| 7 | 85·00 | 86·00 | −1·00 | −2·75 | 28·00 | −1·75 | III |
| 8 | 85·00 | 85·00 | 0·00 | −1·75 | 28·00 | −0·75 | III |
| 9 | 84·00 | 84·00 | 0·00 | −1·25 | 28·00 | −0·25 | III |
| 10 | 83·00 | 84·00 | −1·00 | −1·75 | 28·00 | −0·75 | III |
| 11 | 82·00 | 83·00 | −1·00 | −1·25 | 32·00 | −1·25 | III |
| 12 | 81·5 | 81·5 | 0·00 | 0·00 | 33·00 | −0·25 | III |
| 13 | 82·00 | 83·00 | −1·00 | −1·25 | 32·00 | −1·25 | III |
| 14 | 81·00 | 82·00 | −1·00 | −0·75 | 32·00 | −0·75 | III |
| 15 | 82·00 | 83·00 | −1·00 | −1·25 | 33·00 | −1·50 | III |
| 16 | 83·00 | 83·00 | 0·00 | −0·75 | 32·00 | −0·75 | III |
| 17 | 84·00 | 84·00 | 0·00 | −1·25 | 33·00 | −1·50 | III |
| 18 | 83·00 | 82·00 | 1·00 | 0·25 | 34·00 | −0·25 | III |
| 19 | 82·00 | 84·00 | −2·00 | −2·25 | 32·00 | −2·25 | III |
| 20 | 83·00 | 85·00 | −2·00 | −2·75 | 32·00 | −2·75 | III |

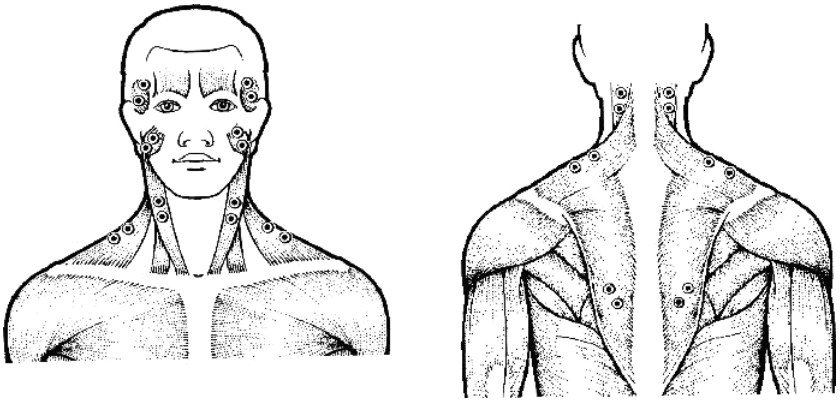

**Fig. 2.** Muscular areas for electrodes placement.

*sEMG recordings*

The study was performed using a Key-Win 2·0 surface electromyograph* with disposable electrodes (DUO F3010 bipolar – 10 mm, Ag–AgCl, lithium chloride gel, unit distance 22 mm).† The Key-Win 2·0 is a sixty channel electromyograph with a 15–430 Hz band-pass filter, containing a special 60 Hz notch filter to eliminate, from the recording environment, all the electrical noises that exceed the capabilities of the common mode rejection scheme. All monitoring was performed with the patients in a standing position. The subjects were asked to have a comfortable posture, to relax their arms, and to look straight ahead and make no head or body movements during the test. The electrodes, which determine the quality of the recordings, were placed according to the electrode atlas of Cram and Kasman (12) (Fig. 2). Before the electrodes were applied, the skin was cleaned with alcohol. The sEMG activity of seven muscles was studied bilaterally (i) with the mandible at rest position, (ii) during MVC. The rest

*Biotronic s.r.l., S. Benedetto Tronto, Ascoli Piceno, Italy.
†LTT FIAB Vicchio, Firenze, Italy.

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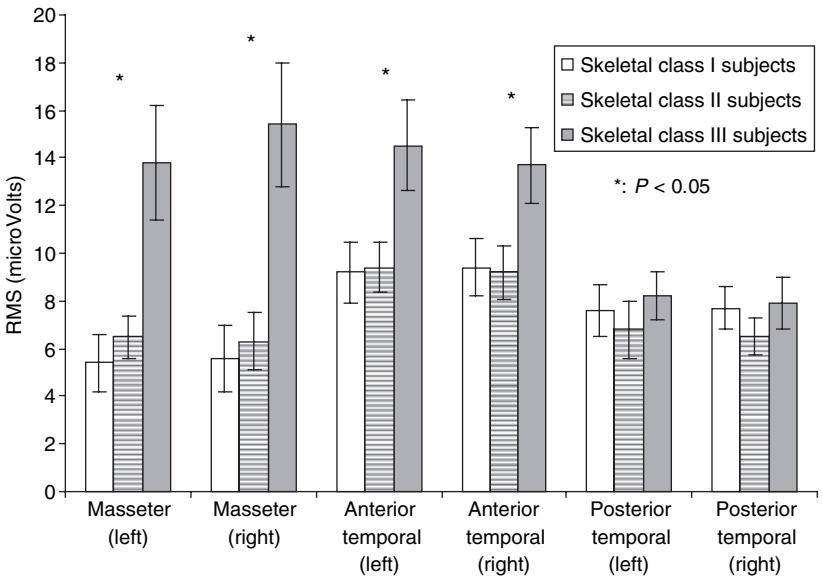


**Fig. 3.** sEMG activity of masticatory muscles at mandibular rest position (Mean and Standard deviations of RMS values in μV).

position was based on verbal instructions to swallow, lick the lips, and say 'Mississippi', according to a recent study which investigated a reliable mandibular rest position (13).

For the MVC recording, the subjects were instructed to close their jaws in habitual occlusion as forceful as possible. Movement patterns were conducted three times to ascertain the stability according to the protocol developed by Donaldson and Donaldson (14). The tracings of the first movements were eliminated as the 'learning' sequence, because they were frequently observed as dissimilar to the other two repetitions. In the single subject, all sEMG data were the arithmetic means of the last two sEMG recordings. The sEMG recordings were randomly performed (i.e. some patients started with mandibular rest position and others started with MVC).

The muscular areas tested were right masseter area, left masseter area, right anterior temporal area, left anterior temporal area, right posterior temporal area and left posterior temporal area as masticatory muscles; right SCM area, left SCM area, right posterior cervicals area and left posterior cervicals area, as neck muscles; right upper trapezius area, left upper trapezius area, right lower trapezius area and left lower trapezius area as trunk muscles. The sEMG recording time for each analysis was at least 15 s, and the values were expressed in microvolts per second (μV $s^{-1}$) (15).

[‡]SPSS, Inc., Chicago, IL, USA.

*Data analysis*

All statistical analyses were performed using the Statistical Package for Social Sciences program.[†] The sEMG data were tested for normal distribution using the Kolmorogov–Smirnov test. The distribution resulted normal and the significance of differences was tested using parametric tests.

For each muscle, the sEMG activity was expressed as mean and s.d. For each condition (mandibular rest position and MVC) and each muscle, a one way ANOVA test and *post hoc* evaluations were employed to test the levels of significance in the differences among the three study groups. The 0·05 level was used to denote statistical significance.

## Results

When the errors in the landmark localization were evaluated on lateral skull radiographs, the differences between the two measurements resulted <5% of biological variance in the whole sample.

Figures 3 and 4 show the sEMG activities of masticatory muscles (mean and s.d.), at mandibular rest position and during MVC, respectively. Figures 5 and 6 show the sEMG activity of the neck and trunk muscles (mean and s.d.), at mandibular rest position and during MVC, respectively.

At mandibular rest position, skeletal class III subjects showed a significantly higher sEMG activity of the

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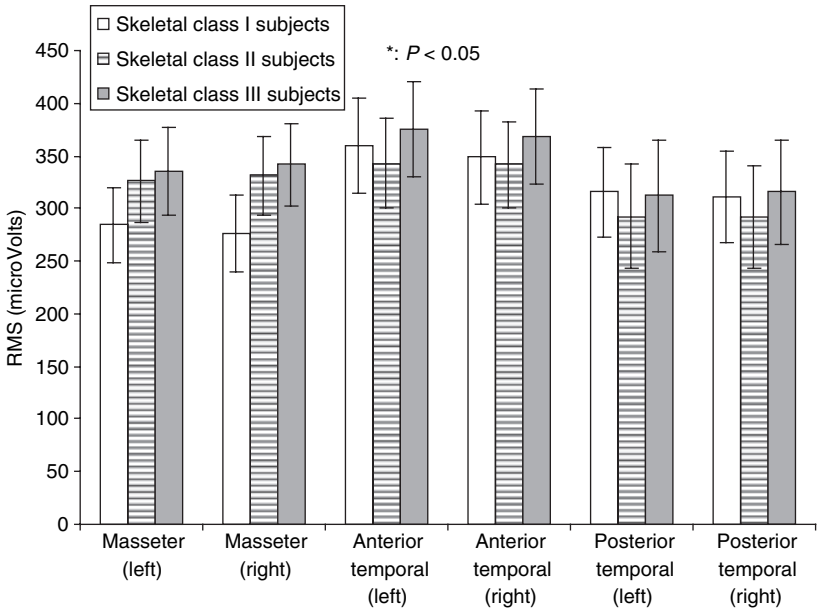


**Fig. 4.** sEMG activity of masticatory muscles during maximal voluntary clenching (Mean and Standard deviations of RMS values in μV).

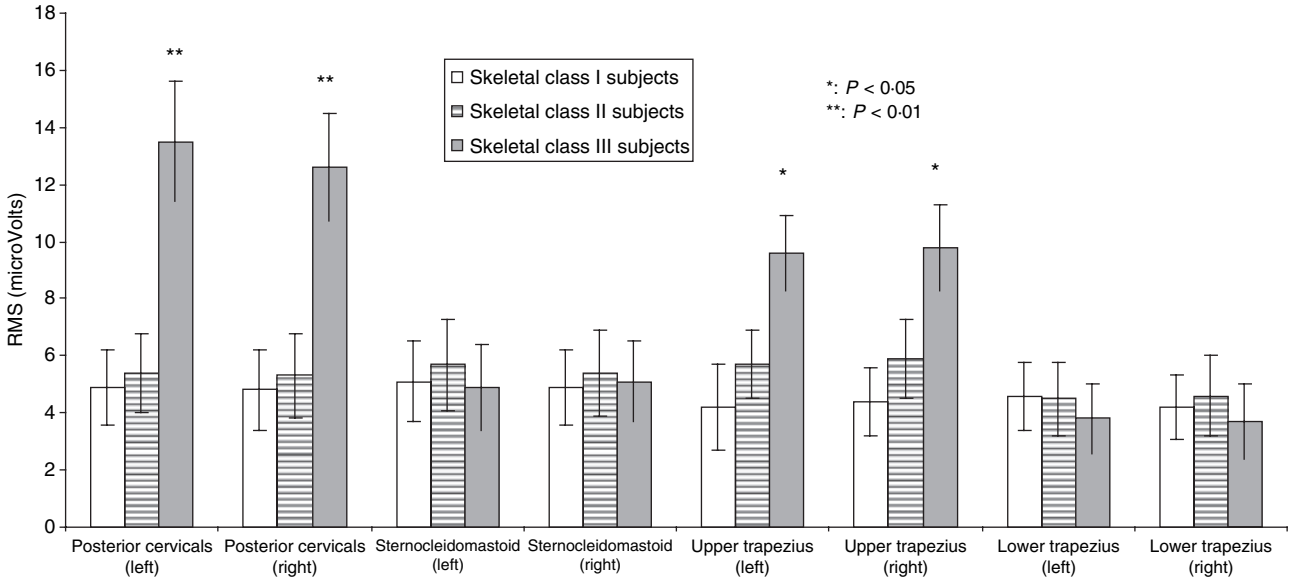


**Fig. 5.** sEMG activity of neck and trunk muscles at mandibular rest position (Mean and Standard deviations of RMS values in μV).

masseter and the anterior temporal, than the other two groups (Fig. 3). However, during MVC, no significant difference was observed among the three study groups (Fig. 4).

Significant differences among the three study groups were observed in the neck and trunk muscles, both at mandibular rest position (Fig. 5) and during MVC (Fig. 6).

The sEMG activity of the posterior cervicals and the upper trapezius in skeletal class III subjects was significantly higher than the other two groups in all the considered conditions (Figs 5 and 6).

## Discussion

### *Masticatory muscles*

The existence of different sEMG patterns activities in subjects with different skeletal facial type was previously observed by Miralles *et al.* (1) for the masticatory

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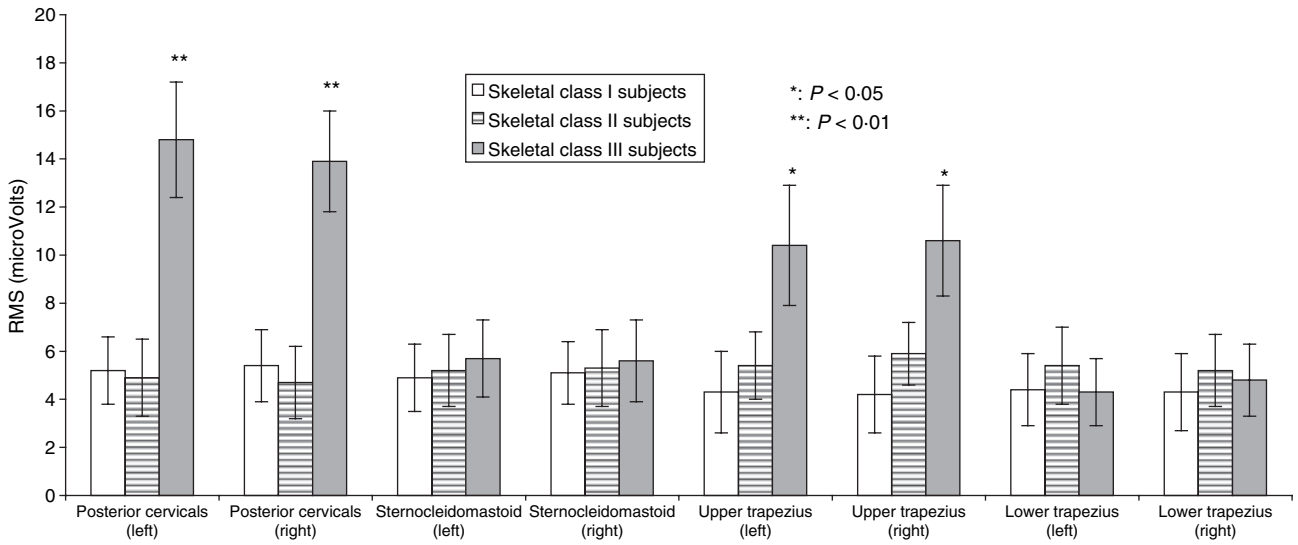


**Fig. 6.** sEMG activity of neck and trunk muscles during maximal voluntary clenching (Mean and Standard deviations of RMS values in μV).

muscles. Our findings seem to be in accord with Miralles *et al.* (1), because skeletal class III subjects showed a significantly higher sEMG activity of masticatory muscles at mandibular rest position, than skeletal class II and class I subjects. Miralles *et al.* (1) underlined the importance of the position and the rotation of the two jaws in class III subjects, as factors capable to determine a change in muscular force axis; class III subjects can show an increase in the vertical component, and a consequent higher stimulation of neuromuscular spindles of the jaw elevator muscles.

In addition, according to Miralles *et al.* (1), we found no significant difference in the sEMG activity of masticatory muscles during MVC among the three study groups.

Recently, a case report was also presented in the literature (16), in which it was observed a correlation between masticatory muscle activity and facial morphology; in this case-report, some differences in the sEMG activity of temporal and masseter muscles were observed before and after a surgical orthodontic treatment in an adult patient with anterior cross-bite and skeletal class III. However, a case report can only be an example because it is not a valid support of the existence of a correlation between masticatory muscle activity and facial morphology. In addition, in another study, an association between craniofacial morphology and temporo-mandibular joint disorders was found (17).

*Neck and trunk muscles*

In the two considered conditions, skeletal class III subjects showed a significantly higher sEMG activity of posterior cervicals and upper trapezius compared with the other two groups.

These observations seem to confirm the existence of neuronal connections between neck and trunk muscles and facial morphology. However, no firm conclusion about this can be given because of the cross-sectional method of this investigation.

In a previous study (18), we observed that skeletal class III children showed a significantly lower cervical lordosis angle and a significantly straighter cervical lower tract of column than skeletal class I and class II children.

According to these findings, skeletal class III subjects seem to show higher sEMG pattern activity of neck and trunk muscles and a straighter cervical column (18) than skeletal class II and class I subjects.

As suggested by several researchers (19), the movement of the lower cervical tract is linked to that of the trunk segment; so, it seems reasonable to hypothize that these two areas (i.e. posterior cervicals and upper trapezius muscles) can be affected, in their own

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function, by the same factors, for example, by facial morphology, as observed in this study. However, Solow and Sonnesen (3) did not find any association between malocclusion and cervical posture, but only between crowding and cervical posture.

In these associations, muscular-neural network seems to play an important role; several researchers (20, 21); for example, underscored that it is responsible for some of common symptoms of the masticatory and/or cervical spine disorders.

Miralles *et al*. (21) also showed, in a group of 15 healthy subjects, a significant increase in the sEMG activity of the neck muscles when varying the vertical dimension every few millimetres from habitual occlusion to 45 mm of jaw opening. Their work confirmed the existence of connections between the occlusion and fusimotor muscle spindle system of the dorsal and neck muscles. Visscher *et al*. (22) also supported these findings from a clinical point of view by showing that the prevalence of cervical spine pain [assessed using oral history and dynamic/static testing with a Visual Analogue Scale (VAS)], was higher in a group of craniomandibular pain patients than in a group of subjects without craniomandibular pain. This was explained by neurophysiological principles of convergence.

## Limits of the study

A limit of this study was that vertical differences (mandibular divergence) were not considered for the association with the sEMG activity of the muscles, although a little correction was made on the ANB angle. This limit was due to the small size of the sample, because the differentiation of sub-groups can reduce the power of statistical analyses.

In addition, females and males were not divided, although a difference in the sEMG pattern activity probably exists; this was done to avoid the decrease of statistical power, but this classification will be the aim of a future investigation, which will evaluate both males and females in different groups. In this study, only adult females were considered to avoid errors due to the sex dimorphism.

## Conclusions

In conclusion, our findings seem to support the concept that morphological aspect of face influences the sEMG pattern activity of masticatory, as well as the neck and trunk muscles. In particular, in a Caucasian adult females sample, skeletal class III subjects showed a significantly higher sEMG pattern activity at mandibular rest position of the posterior cervicals and upper trapezius muscles, compared with skeletal classes I and II subjects. Future longitudinal evaluations will contribute to clarify the complex mechanism.

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